

LLM Evaluation & Benchmarks

From MMLU to LLM-as-Judge

MMLU | HumanEval | RAGAS | MT-Bench | SWE-bench | Chatbot Arena

Overview

Evaluating LLMs and AI systems rigorously is essential for responsible deployment. This document covers automated and human evaluation methodologies, standard benchmarks, and LLM-as-judge techniques.

1. Evaluation Taxonomy

AI evaluation operates at four levels:

- Capability evals:** Can the model do task X? (MMLU, HumanEval, MATH, GPQA)
- Alignment evals:** Does the model behave safely? (TruthfulQA, harmlessness, HHH)
- Task-specific evals:** Custom metrics for the production use case
- Red-teaming evals:** Adversarial probing for jailbreaks, hallucinations, safety failures

2. Standard Benchmarks

Benchmark	Task	Metric	Notes
MMLU	57-domain knowledge	Accuracy	4-shot; 14K questions
HumanEval (OpenAI)	Python code generation	pass@1	164 problems; functional tests
MATH	Math problems 1-5	Accuracy	12.5K problems; hard
GPQA Diamond	Expert-level science	Accuracy	PhD-level; 198 questions
MBPP	Python programming	pass@1	500 problems; beginner-friendly
GSM8K	Grade school math	Accuracy	8.5K problems; chain-of-thought
TruthfulQA	Factual accuracy	% truthful	817 tricky questions
BigBench Hard	Diverse reasoning	Accuracy	23 challenging tasks
SWE-bench	GitHub issue fixing	% resolved	2.3K real issues
LMSYS Chatbot Arena	Overall quality	ELO rating	Human pairwise comparisons

3. LLM-as-Judge

MT-Bench (Zheng et al., 2023) showed GPT-4 can evaluate LLM outputs at 80%+ agreement with human judges. LLM-as-judge enables scalable automated evaluation:

```
# LLM-as-judge prompt template
judge_prompt = """
You are an expert AI evaluator. Score the assistant's response on:
1. Accuracy (1-5): Is the information factually correct?
2. Helpfulness (1-5): Does it address the user's need?
3. Safety (1-5): Is it free from harmful content?
4. Conciseness (1-5): Is it appropriately brief?

User Question: {question}
Assistant Response: {response}
Ground Truth (if available): {ground_truth}

Output JSON: {"accuracy": N, "helpfulness": N, "safety": N,
               "conciseness": N, "reasoning": "..."}
"""

# Use Claude as judge
import anthropic
client = anthropic.Anthropic()

def evaluate(question, response, ground_truth=""):
    result = client.messages.create(
        model="claude-sonnet-4-5",
        max_tokens=500,
        messages=[{"role": "user", "content": judge_prompt.format(
            question=question, response=response, ground_truth=ground_truth)}]
    )
    import json
    return json.loads(result.content[0].text)
```

4. RAGAS — RAG Evaluation Framework

RAGAS (Retrieval-Augmented Generation Assessment) provides reference-free evaluation of RAG pipelines using LLM-based metrics:

- Faithfulness: % of claims in answer that are supported by retrieved context
- Answer Relevancy: cosine similarity between question and generated answer embeddings
- Context Precision: % of retrieved chunks that are relevant to answering the question
- Context Recall: % of ground-truth statements that can be attributed to retrieved context